

TopoGym: Library Overview and Environment Specifications

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Scope. This document specifies the TopoGym environment library in depth: the base benchmark (**GridWorld2D**), its textured variant (**GridWorld2D-Texture**), and its topological variant (**GridWorld2D-Top**, non-trivial global topology via edge identifications). Together the three variants form one benchmark, **TopoGym-v1**: the versioned, pinned registry across all variants under the universal action and observation spaces of Section 1.9 — Plain, Texture, and Top are reporting slices of a single suite, and the version pins the environment list (future families extend to v2 without altering v1 entries). The experiment program that consumes TopoGym-v1 is specified in `papers/topoexplore/experiment_design.tex` (campaign X1); this document is the authority on the environments themselves. GridWorld2D is specified fully here; the Texture and Top variants are outlined, with their detailed scenario and construction specifications to follow.

1 GridWorld2D

A registry of named environments backed by one parametric generator. Everything is generated from a frozen configuration and a seed; nothing is hand-placed. The design goals: ground truth matches the theory’s objects exactly, so mechanism predictions and certificates are checkable programmatically; and the interface follows existing gym-library conventions — stable named environments with seed variation and a few modifiable parts — so the benchmark is directly usable by others.

1.1 World model

A `size × size` grid of cells, partitioned into free and obstacle cells. The agent occupies one free cell; actions are {up, down, left, right}; moving into an obstacle leaves the agent in place. Standing conventions, inherited from the theory document and enforced at generation time:

- the exterior free space (the component containing the start cell) is 4-connected;
- obstacles are well-composed: no 2×2 window contains exactly a diagonal pair of obstacle cells (no diagonal pinch);
- every door has width one: a free cell with obstacle cells on two opposite sides and free cells on the other two;
- a chamber is an obstacle wall enclosing free interior once its doors are sealed; a decoy is a sealed ring enclosing no reachable free cell — its interior is unvisitable by construction, which is what makes it a decoy from the archive’s perspective.

1.2 Registry: named environments (the primary interface)

Following the convention of existing gym libraries, TopoGym’s canonical interface is a registry of named, pre-defined environments: `gym.make("TopoGym/{Family}-{size}-v0", seed= $n$ )`. Each registry entry is a frozen configuration of the underlying generator; the seed drives layout variation (placement, door positions, shape assignment) within that configuration; and a small set of documented parts remains modifiable as `make` kwargs (`p_slip`, `reward_mode`, and per-family knobs noted below). The benchmark is the registry crossed with the published seed lists.

Family	Description	Sizes / variants
Dilution-{50,200}	one chamber, no decoys	—
Chambers2-{50,100,200,400}	two chambers, fixed geometry; the world-scaling family	—
ChamberCount{1,2,4,8}-200	k separated chambers	kwargs: none
Decoys{0,1,2,4,8}-50	one chamber among k_d sealed decoys	kwargs: <code>decoy_side</code>
Shape{Sq,Ci,Tr,St}-50	area-matched chamber shapes	—
Nested{1,2,3}-50	sequentially nested shells	—
GiveUp{1,2,4}-50	door behind a dead-end corridor of the given length	—
Bottleneck{3,6}-100	tree of rooms joined by width-1 corridors of the given length	kwargs: <code>rooms</code>
Maze-{50,100}	seeded perfect maze (uniform spanning tree over cells)	kwargs: <code>braid</code>

Registry ids are stable across releases; new families extend the registry rather than altering existing entries.

1.3 Generator API (advanced)

Under the registry sits the full parametric generator, available to power users and used to define registry entries. Its configuration schema:

- **size** — grid side length; any integer accepted.
- **chamber_count**, **decoy_count**; **chamber_side**, **decoy_side**; **chamber_shape**, **decoy_shape** ({square, circle, triangle, star, mixed}, area-matched at equal **side** so shape is never confounded with size).
- **min_sep** — minimum pairwise wall separation; configurations claiming the one-entry-per-rollout property must satisfy $\text{min_sep} > L$ (rollout horizon), recorded per configuration in the manifest.
- **doors_per_chamber** (default 1); **door_width** fixed at 1 and validated; **door_corridor_len** (default 0), controlling entry probability q .
- **mode** — open, nested, corridor, or maze (Section 1.4).
- **p_slip**, **reward_mode**, **seed**.

A configuration serializes to the canonical string `TG-GridWorld2D-S{size}-C{c}-D{d}-cs{n}-ds{n}-sep{n}-shp{shape}` which is the run-log key and the manifest row; registry ids are aliases for canonical strings. Custom configurations built through this API pass the same validity checks and ground-truth generation as registry entries, so results on them remain comparable in kind, but only registry entries belong to the published benchmark.

1.4 Modes

- **open**: chambers and decoys placed in a free field by seeded rejection sampling subject to `min_sep`. The direct realization of the theory’s objects: dilution scales with `size`, discrimination with `decoy_count`, parallel entry requires the separation condition. Generation: sample footprints uniformly, reject on overlap or separation violation, carve walls and doors, validate.
- **nested**: `depth` concentric shells around one innermost chamber, one door per shell, doors angularly offset so entry forces traversing shells in order. The sequential regime where the parallel-chamber separation deliberately fails, and where learned-persistence baselines are predicted to win. Bars are born sequentially: entering shell j first exposes shell $j+1$ ’s cycle. Validity: shell spacing ≥ 2 ; depth $\times$ spacing bounded by half the side.
- **corridor**: rooms joined by width-1 corridors of parameterized length, the room graph constrained to a tree. The tree constraint is load-bearing: a cyclic room graph would enclose wall blocks and manufacture decoy-like H_1 bars, while a tree keeps free space simply connected — zero environmental homology signal, only transient exploration pockets. The bottleneck regime; corridor length is the bottleneck-depth difficulty axis. Ground truth additionally exposes the room graph and corridor cells.
- **maze**: seeded perfect-maze carving (a uniform spanning tree over cells), the maximal-corridor-density cousin of **corridor** mode — simply connected at `braid` = 0. The `braid` parameter opens loops with the given density; each opened loop encloses wall and therefore creates an incidental H_1 class, so braided mazes interpolate between the bottleneck regime and the discrimination regime.

1.5 Validity checking and manifests

The generator validates before building and rejects with a reason: packing feasibility (footprint plus separations fits, by a conservative area bound and then detection of rejection-sampling failure); exterior 4-connectivity after carving; well-composedness; door width; separation claims. The manifest tool emits a CSV of (registry id, canonical configuration, validity, satisfied assumptions) covering the registry — and, for power users, any custom configuration set. The published benchmark is the registry manifest plus fixed seed lists — tuning seeds and evaluation seeds, identical environments, disjoint seeds — as specified in the experiment design.

1.6 Ground truth, certification, and instrumentation

Read-only accessors per instance: the wall mask; the chamber list with interiors $I(W_i)$, perimeters P_i , door cells $D(W_i)$; the decoy list with representative rims; per-cell annotations (chamber and door membership; full-graph Ollivier–Ricci curvature, computed once by the generator); and exact snapshot save/restore. A certification checker validates a reported class list against the decoy ground truth and each certificate (every rim action tried $u_{\max}$ times without discovery, or explicit interior filling) directly from the run log. The shared recorder writes the full metric catalogue of the experiment design to Parquet at four granularities (per step, per epoch, per training run, per evaluation run). Registration-time unit tests assert the standing conventions on every generated instance.

1.7 Complex backends (all variants)

Every environment exposes a **complex** configuration field selecting the geometric substrate used for free-space homology:

- **cubical** (default) — the glued cubical cell complex of the free cells; homology is computed by GUDHI on the order complex of its face poset (the barycentric subdivision), which is robust to every edge identification of Section 3. This backend is the certification source of truth: the metadata’s Betti numbers are computed here and cross-checked against the analytic expectation at generation time.
- **rips** — a Vietoris–Rips complex built by GUDHI on the centers of the free cells at a fixed scale $t \in (\sqrt{2}, 2)$, under the quotient metric induced by the base’s identification group (wrap translates and flip glide reflections). Orthogonal and diagonal neighbors connect, cells separated by a width-one wall (ambient distance 2) do not, so obstacles produce classes exactly as in the cubical complex; the backend reports dimensions 0 and 1, and the library’s tests assert agreement with the certified cubical numbers on every base.

The backend governs the environment-side homology computations (**free_betti**, **visited_betti**, **observed_betti**) and is recorded in the metadata; certification always comes from the cubical complex.

Determinism. The library guarantees end-to-end determinism up to seeds: (configuration, seed) fixes the generated layout and its certified metadata byte-for-byte — including everything computed through GUDHI, since homology is exact linear algebra over $\mathbb{F}_p$ with no randomness — and (environment, reset seed, action sequence) fixes the episode, including **p_slip** resampling, which draws from the episode’s seeded generator. Internal iteration orders are sorted so no result depends on interpreter hash state; a cross-process test asserts that two fresh interpreters produce identical layouts, metadata, and rollouts.

1.8 Reward modes (all variants)

Every benchmark variant (GridWorld2D, -Texture, -Top) supports a **reward_mode** configuration field:

- **none** (default) — no extrinsic reward; the environment is a pure-exploration benchmark, and event-based metrics (entry times, coverage) carry the evaluation.
- **sparse** — +1 terminal reward on reaching a target cell placed inside a designated chamber (in Texture scenarios, the treasure-chest cell). Placement inside a chamber makes steps-to-first-reward coincide with steps-to-first-entry, so the reward metric stays continuous with the exploration metric.
- **coverage** — +1 for each cell visited for the first time: a dense pure-exploration reward for methods that require a reward signal.
- **deceptive** — the **sparse** target plus a distractor shaping gradient leading away from the target’s door: the reward-deception regime of the novelty-search literature, letting reward deception be measured rather than only structural deception.

Ground truth exposes the reward cells and (for **deceptive**) the shaping field. Reward modes exist for community use, for training reward-consuming components (the hole-value and frontier heads, the planner’s exploitation term), and for deception studies; the TopoExplore experiment program runs **none** throughout to keep an extra variable out of its main results.

1.9 Universal action and observation spaces

Both spaces are identical across all variants (GridWorld2D, -Texture, -Top); an agent written against them runs on every environment in the library unchanged.

Action space. `Discrete(4)`: up, down, left, right. Moving into an obstacle leaves the agent in place. With `p_slip` > 0 the executed action is resampled uniformly with that probability. In GridWorld2D-Top, crossing an identified edge applies the identification map (coordinate and orientation reversal where the identification is flipped) — the action space itself is unchanged. Family-specific cell mechanics (hazard cells that reset the episode, wormhole cells that teleport; Section 2) are documented with the family and never alter the action set.

Observation space. The canonical observation is a fixed vector: the agent’s integer cell coordinates (x, y) followed by a texture block $t \in [0, 1]^{16}$. Slots 0–3 are reserved library-wide for directional blocker adjacency (obstacle to the left, right, above, below of the current cell); slots 4–15 carry per-environment semantic features documented with each family. The texture block is identically zero in GridWorld2D and GridWorld2D-Top; only GridWorld2D-Texture populates it. Optional observation modes for learned baselines — flattened symbolic grid, or an egocentric window of configurable radius with per-cell texture — are available on every variant. Snapshot save/restore is exact (deep copy), and the environment is deterministic given (configuration, seed, action sequence) when `p_slip` = 0.

2 GridWorld2D-Texture

The textured variant (TG-GridWorld2DTexture prefix): the same generator with a per-cell texture channel — semantic labels such as blocker-adjacent (ice), door, ladder, water, and scenario-specific others — exposed in the observation and recorded in ground truth. Texture is a third signal family: local like curvature, but semantic rather than geometric, and only as trustworthy as the environment’s labeling.

Design intent. Texture scenarios are built so the taxonomy’s boundary is visible: texture discriminates flagged structure cheaply where the labeling is faithful, and fails exactly where discrimination requires enclosure rather than adjacency. The flavor example: arctic sailing — most holes are ice-cube decoys whose neighbors carry ice-adjacent texture (avoidable on sight); the target is a treasure chest in an enclosed space reachable through a water-only passage; texture flags ice adjacency and doorways (ice on both sides), while whether the passage leads anywhere remains invisible to any local feature. Scenario suites should include cells where texture is absent or misleading, so the boundary is measured rather than assumed.

Texture encoding. Textures populate the universal observation’s texture block (Section 1.9): slots 0–3 carry directional blocker adjacency (blocker to the left, right, above, below — any combination), and slots 4–15 carry the scenario’s semantic features, documented per environment below.

Canonical environments. Five named environments form the Texture slice of TopoGym-v1:

- **TopoGym/IceShip.** Arctic ship navigation. The world is water (navigable) and ice (blocker); most holes are sealed ice decoys, and a treasure chest sits inside one true chamber. Ice-adjacent water cells carry the directional adjacency flags (ice to the left, right, above, below, or any

combination — a doorway reads as ice on both sides); open-water cells carry the water feature only. Water cells, ice-adjacent or not, are the only navigable cells. Texture makes decoy avoidance cheap on sight; whether a passage leads anywhere remains invisible to texture.

- **TopoGym/Ladders.** Platforms, ladders, and bridges; a gem on the top platform is the target. Rooms (platforms), ladders, and bridges each carry their own semantic feature; vertical progress is only possible on ladder cells, and bridges connect platforms across gaps. The textured version of the bottleneck regime: ladders and bridges are the corridors, and their texture advertises them.
- **TopoGym/BankRobber.** Nested rooms with the money in the center room — the textured version of the nested regime. Door cells and hallway cells carry their own features, so the sequential structure is advertised locally; the ordering constraint (which door must come first) remains global.
- **TopoGym/DontFall.** A large world with a large central hole: stepping onto the drop resets the episode (the hazard mechanic). A few dozen much smaller huts surround the hole, and exactly one contains the ruby. Textures distinguish drop-adjacent squares, regular dirt, hut interiors, and hut doorways. The hazard makes local novelty signals actively dangerous — the most novel direction is the drop — while the huts reproduce the discrimination regime at scale.
- **TopoGym/SpaceWarp.** Four chambers and wormholes: paired teleporter cells placed near doors at a configurable separation (kwarg `warp_sep`) — never directly in front, so every door remains enterable. Stepping onto a wormhole teleports the agent to its partner (the teleport mechanic). The treasure chamber has no door at all: from outside it is indistinguishable from a sealed decoy, and it is enterable only through a wormhole inside another chamber. Texture flags a cell as a wormhole but never which one — all wormholes are purple — so wormholes are recognizable on sight while their destinations must be explored. The deception cuts both ways: a wormhole can be a shortcut or a stranding, and only entering it tells.

SpaceWarp and the certificate (semantics, not soundness). SpaceWarp does not break the certificate — it sharpens what the certificate says. Both certified statements remain true here: the treasure chamber’s enclosing class is genuinely permanent in the spatial complex (its wall has no door, so the class never dies even after wormhole entry archives the interior — the wall stays uncovered and the hugging cycle never bounds), and its gate log is genuinely exhausted (no local entry exists). What the environment changes is the gloss: a decoy certificate certifies no local entry and a permanent spatial H_1 class, not unenterability — with non-local transitions, certified decoys can still be enterable through a wormhole. SpaceWarp is included to make that semantic boundary measurable; the extension that upgrades the certificate from “no local entry” to “no entry” — evaluating gate evidence and the complex on the archive transition graph, where wormhole edges appear once traversed — is future work the environment is built to test.

Status. The scenario constructions above fix the suite; exact sizes, layouts, semantic slot assignments, and generation rules are to be specified in a forthcoming revision of this document.

3 GridWorld2D-Top

The topological variant (TG-GridWorld2DTop prefix): non-trivial global topology via edge identifications of the grid’s boundary. The `topology` field selects the identification:

Value	Identification	Space	$H_1(\cdot; \mathbb{Z})$	$H_1(\cdot; \mathbb{F}_2)$
plane	none (walled boundary)	disk	0	0
cylinder	left–right, straight	cylinder	$\mathbb{Z}$	$\mathbb{F}_2$
mobius	left–right, flipped	Möbius band	$\mathbb{Z}$	$\mathbb{F}_2$
torus	both pairs, straight	torus	$\mathbb{Z}^2$	$\mathbb{F}_2^2$
klein	one straight, one flipped	Klein bottle	$\mathbb{Z} \oplus \mathbb{Z}/2$	$\mathbb{F}_2^2$
rp2	both pairs, flipped	real projective plane	$\mathbb{Z}/2$	$\mathbb{F}_2$

Movement wraps across identified edges, with coordinate reversal where the identification is flipped; obstacles and chambers can be placed on top of any topology. Note the torsion: over $\mathbb{Z}$ the Klein bottle and $\mathbb{RP}^2$ carry $\mathbb{Z}/2$ classes that integer rank alone misses, while $\mathbb{F}_2$ coefficients see them — one reason the library computes homology over $\mathbb{F}_2$.

Why it matters. These spaces are locally flat everywhere: no local signal — curvature, texture, novelty gradient, count statistics — distinguishes a Möbius band from a plane at any single cell, because every neighborhood is Euclidean. The global structure is visible only to signals that aggregate globally, and the ambient H_1 is non-trivial with no obstacles at all. This is the regime designed to break methods that rely on local exploration gradients, and the cleanest possible separation between local and global signal families.

Canonical layout. The Top environments share one scenario shape: chambers placed near the corners of the large fundamental square, exactly one holding the treasure. The square’s sides are the edges of the identification diagram (the fundamental polygon), so the corner regions meet across the identified edges — on the torus all four corners are a single point of the quotient; on the Klein bottle and $\mathbb{RP}^2$ they meet in pairs with orientation reversal. What appears locally as four maximally separated chambers is globally a tight cluster reachable across the edges; an agent that carries only local structure treats the corners as far apart, while the quotient makes them neighbors.

Status. Identification-aware metric and complex construction (the Rips complex on the quotient metric), the interaction of wrap-around with the archive and with return-by-replay, detection of the ambient classes versus feature classes, and the per-topology registry entries are to be specified in a forthcoming revision of this document.